

NANYANG TECHNOLOGICAL UNIVERSITY

MIDTERM II

MH2100 – Calculus III

October 2022

TIME ALLOWED: 90 minutes

INSTRUCTIONS TO CANDIDATES

1. **DO NOT TURN OVER PAPER UNTIL INSTRUCTED.**
2. This midterm paper contains **THREE (3)** questions.
3. Answer **ALL** questions. The marks for each question are indicated at the beginning of each question.
4. Candidates should write their answers in the separate answer book provided.
5. **ONE** two-sided A4 piece of paper with handwritten notes is permitted.

QUESTION 1.**(16 marks)**

- (i) [5 marks] Find the *maximum* and *minimum* values of xy satisfying the condition that $3x^2 + 2y^2 = 36$. Also give the points (x, y) where the *maximum* and *minimum* values are achieved.
- (ii) [3 marks] Evaluate the iterated integral.

$$\int_{-1}^1 \int_0^1 2ye^x \, dx \, dy.$$

- (iii) [4 marks] By reversing the order of integration, evaluate the integral

$$\int_0^{\sqrt{\pi/2}} \int_y^{\sqrt{\pi/2}} 2 \cos(x^2) \, dx \, dy.$$

- (iv) [4 marks] Find the area of the region D (in the xy -plane) bounded by the cardioid expressed in polar coordinates as

$$r = 1 - \sin \theta; \quad \text{and} \quad 0 \leq \theta \leq 2\pi.$$

Hint: the formula $\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}$ may be useful.

QUESTION 2.**(17 marks)**

- (i) [4 marks] Use cylindrical coordinates to evaluate

$$\iiint_E \sqrt{x^2 + y^2} dV,$$

where E is the solid bounded by the cylinder $x^2 + y^2 = 1$ and the planes $z = 1$ and $z = 2$.

- (ii) Consider the transformation
- $u = x + 4y$
- and
- $v = y - x$
- .

- (a) [3 marks] Calculate the Jacobian
- $\frac{\partial(u, v)}{\partial(x, y)}$
- .

- (b) [5 marks] Use the transformation to evaluate the iterated integral

$$\int_0^1 \int_y^{5-4y} (x + 4y)e^{y-x} dx dy.$$

Hint: you can use $\int x e^{ax} dx = \frac{e^{ax}(ax - 1)}{a^2} + C$.

- (iii) [5 marks] Use spherical coordinates to find the volume of the region inside the sphere
- $x^2 + y^2 + z^2 = 4$
- and outside the cone
- $z^2 = x^2 + y^2$
- . That is, the volume of the region
- E
- given by

$$E = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 4 \text{ and } z^2 \leq x^2 + y^2\}.$$

QUESTION 3.**(17 marks)**

- (i) [4 marks] Let C be the curve parametrised by $x = 3t$ and $y = t^3$ where $0 \leq t \leq 1$. Evaluate

$$\int_C (x^3 + y) ds.$$

- (ii) [4 marks] Find the work done by the vector field $\mathbf{F} = \langle x^3 - y^3, xy^2 \rangle$ along the curve parametrised by $x = t^2$ and $y = t^3$ from $t = -1$ to $t = 0$.
- (iii) [4 marks] Determine if the vector field $\mathbf{F} = \langle 3x^2y^2 + e^x, 2x^3y + \sin(y) \rangle$ is conservative. If so, find a potential function f for $\mathbf{F}$.
- (iv) [5 marks] Let C be part of the curve (in the xy -plane) described by polar coordinates as $r = \sin(\theta)$ where $0 \leq \theta \leq \pi$. Find the length of C .